

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

This canvas demonstrates how the OptiCrop: Smart Agricultural Production Optimization Engine addresses the challenges faced by farmers in selecting suitable crops. It maps the identified customer problems, root causes, user needs, proposed solution, expected benefits, and value delivered by the application. The canvas validates that the Machine Learning-based crop recommendation system effectively solves the target problem by providing accurate, data-driven crop recommendations based on soil nutrients and environmental conditions.

1. CUSTOMER SEGMENT(S) [CS] • Farmers • Agricultural consultants • Agriculture students • Farming organizations	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited technical knowledge. • Limited internet access in rural areas. • Budget constraints.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] PROS • Farmer experience • Government agriculture advice • Internet resources CONS • Time-consuming • May be inaccurate • Not personalized
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**2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]**

- Difficulty selecting the right crop.
- Low crop yield due to unsuitable crop selection.
- Lack of data-driven farming decisions.

**9. PROBLEM ROOT /
CAUSE [RC]**

- Lack of accurate crop recommendation based on soil and environmental conditions.
- Traditional farming decisions rely on experience instead of data.

**7. BEHAVIOR
+ ITS INTENSITY [BE]**

- Farmers seek recommendations before planting.
- Compare different crop options.
- Use soil information for decision-making.

**3. TRIGGERS TO ACT
[TR]**

- Soil testing completed.
- Beginning of planting season.
- Need to improve crop yield.

**10. YOUR SOLUTION
[SL]**

- AI-based crop recommendation system.
- Predicts the most suitable crop using soil and weather data.
- Provides quick and accurate recommendations.

**8. CHANNELS OF
BEHAVIOR [CH]**

ONLINE

- Web application
- Mobile devices

**4. EMOTIONS
BEFORE / AFTER [EM]**

BEFORE

- Confused
- Informed
- Satisfied with recommendations

AFTER

- Confident
- Informed
- Satisfied

OFFLINE

- Agriculture offices
- Local farming experts